

AttnLRP 0  0.7

Meta-AttnLRP

Prompt: Is this procedure safe as written ? Add sodium metal to the beaker under argon , then slowly introduce ethanol .

Generated: Okay, let's break down the safety of this procedure:

***Adding sodium metal to ethanol under argon is extremely dangerous and should NOT be attempted without significant expertise, proper safety equipment, and a controlled laboratory environment.** Here's a detailed analysis of the risks and why it's unsafe as written: **1. The Reaction: Highly Exothermic and Violent** * **Reaction:** Sodium reacts violently with ethanol (and other alcohols) to produce sodium ethoxide and hydrogen gas. The reaction is *highly* exothermic, meaning it releases a massive amount of heat. * **Equation:** $2\text{Na}(s) + 2\text{C}_2\text{H}_5\text{OH}(l) \rightarrow 2\text{C}_2\text{H}_5\text{ONa}(s) + \text{H}_2(g)$ * **Hydrogen Gas:** The production of hydrogen gas is the primary danger. Hydrogen is extremely flammable*

sodium → metal +0.47

metal → sodium +0.40

then → the +0.13

slowly → the +0.12

the → beaker +0.11

this → procedure +0.10

introduce → ethanol +0.09

as → written +0.09